

Tuning curve dynamics in a bump attractor model of the auditory cortex

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1 Introduction

Ring attractor networks are a class of continuous attractor models with recurrent symmetric connections used in theoretical neuroscience. These networks are characterized by a continuous attractor manifold, where the network’s activity forms a local "bump" corresponding to peak firing activity within part of the ring, which stabilizes to a final location despite arbitrary initial states. Each point on the ring represents a particular value of a continuous variable, such as the orientation of a stimulus, or the position of a head-direction signal. The recurrent structure allows the network to exhibit tuning in the form of a stable bump, which can encode information in the absence of continuous external input. Due to this nature, they have been proposed to underlie cognitive functions dealing with working memory, and experimental data from the head direction network and visual tuning orientation have led to the development accurate models of tuning in HD and V1 cells. However, in the auditory cortex, these models have remained largely underutilized. However, many features of auditory input and perception are periodic in nature, providing reasonable basis for ring attractor representations of neural activity.

The auditory cortex is involved in processing sounds across a broad range of frequencies. Neurons in the auditory cortex are tuned to specific frequencies, and their response properties are often organized spatially within the cortex. In this study, we propose that in the auditory cortex, where harmonics play a larger role, multiple bumps in single ring can capture some of the tuning dynamics of overlapping acoustic input. To capture this in the model, we attempt to introduce the formation of multiple bumps to capture different frequencies within a harmonic input. While the model has yet to produce multiple bumps, the results reflect a broader tuning curve with more overall neuron activation, indicating that the auditory cortex may share a basic winner-takes-all regime with other ring attractor networks.

2 Methods

2.1 Model Structure

The ring attractor structure in this study is modeled using leaky integrate-and-fire neurons, modeling an excitatory ring. The membrane potential of neuron k, j within a barreloid/barrel system is governed by a reduced dimensional integrate-and-fire model of a cortical cell:

$$\frac{dV^{k,j}}{dt} = -g(V^{k,j} - V_{\text{rest}}) + I^{k,j}(t), \quad (1)$$

$V^{k,j}$ is the non-dimensional membrane potential, $g = 0.05 \text{ ms}^{-1}$ is the leak conductance, and $I^{k,j}(t)$ is the synaptic current (in ms^{-1}). $V_{\text{rest}} = 0$ is the resting potential, and a spike is recorded when $V^{k,j} \rightarrow 1^-$, at which point $V^{k,j}$ is instantaneously reset to V_{rest} . A refractory period is simulated by holding $V^{k,j}$ at V_{rest} for 2 ms following a spike. The integrate-and-fire equation has a membrane time constant of 20 ms.

A spike of a neuron presynaptic to neuron k, j leads to a jump in $I^{k,j}(t)$ followed by exponential decay, after a manually imposed synaptic latency. Let $n^{k,j}$ denote the total number of presynaptic spikes that impinged upon neuron k, j during a trial. If the r^{th} presynaptic spike occurs at time $t_r^{k,j}$, and $m \in \{\text{tc}, \text{fs}\}$ is the type of the presynaptic neuron, the current $I^{k,j}(t)$ induced in neuron k, j at time t is given by the following:

$$i_r^{k,j}(t) = \begin{cases} 0 & t < t_r^{k,j} + d_m^k, \\ A_m^k e^{-\alpha_m^k(t - t_r^{k,j} - d_m^k)} & t \geq t_r^{k,j} + d_m^k. \end{cases} \quad (2)$$

$$I^{k,j}(t) = \sum_{r=1}^{n^{k,j}} i_r^{k,j}(t). \quad (3)$$

Simulations are carried out using RK4 with a time step of 0.1 ms. The network was also tested using a timestep of 0.01 ms, to validate that it is invariant under step size.

In the excitatory ring, connectivity is set by a weights matrix representing the ring of excitatory neurons, with weights being determined by a von Mises distribution. Let the network consist of N neurons arranged on a ring, each with a preferred angle $\theta_i \in [0, 2\pi)$. The synaptic weight from neuron j to neuron i is given by entry $W_{i,j}$ in the weight matrix W . To ensure translation-invariant connectivity that supports bump formation, the weights are defined as a function of the difference between preferred angles:

$$W_{i,j} = (A/N)^{(\kappa \cos(\theta_i - \theta_j))}$$

Where A is the parameter for synaptic strength, κ is the concentration of the von Mises peak, and N represents number of excitatory neurons in the network. This ensures that the network is scale invariant. In initial modeling, we began with global inhibition, first represented by a network of only excitatory neurons, where inhibition is a single value B ;

$$W_{i,j} = (A/N)^{(\kappa \cos(\theta_i - \theta_j))} - B$$

Further development of the model then used a single inhibitory neuron with flat weights, projecting to all other neurons in the model. In the version of the excitatory ring used in this study, inhibition to the ring is global and tuned, while excitatory to inhibitory and inhibitory to inhibitory weights are flat and constant. A complete log-scaled weights matrix with sample network sizes is shown in Fig. 1.

Despite experimental data indicating that the bump is typically Gaussian in shape, our network produces a sharper tuning curve. The introduction of failure rates, a corresponding to the biophysical phenomena where a neuron fails to transmit a signal despite spiking itself, introduces stochasticity into the network, slightly decreasing activity at the peak relative to the tails. Firing rates are plotted as histograms of spike counts, are smoothed using convolution by a Gaussian kernel, using a sliding 50 ms bin.

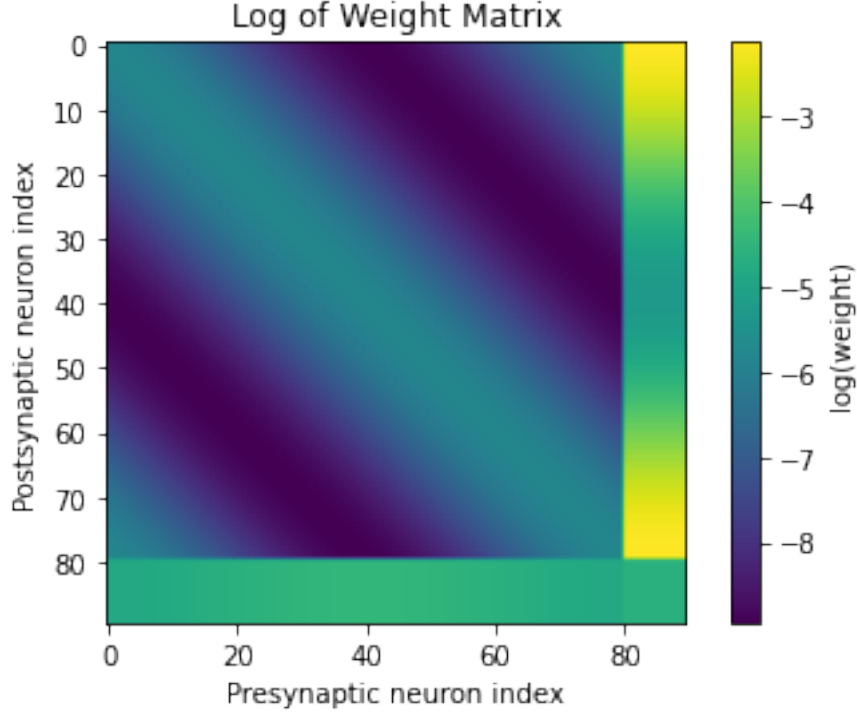


Figure 1A. Log-scaled weights matrix for a network of 40 E and 10 I neurons.

3 Results

3.1 Symmetry Breaking

In this network, all weights are symmetrical. In the symmetric regime, the network’s architecture ensures that the recurrent connectivity is translation-invariant, supporting a continuous manifold of equally stable states corresponding to different positions of a localized bump of activity. However, this symmetry must be broken to encode a specific value in working memory or to respond selectively to external stimuli.

Initial network state is random, and input to the network activates a set of neurons in a location corresponding to a "cue" within the orientation. In this network, only a subset of neurons within the ring corresponding to the bump formation spike after the network settles from an initial state. Spike rasters are displayed in Fig. 2, showing the formation of a self-sustained bump for the active neurons within the first 200ms of the simulation. Figure 3 shows PSTHs of both population of neurons with binsize of 10. Tuning curves indicate that this network stabilizes very quickly into a bump regime, despite arbitrary initial conditions, and even with noise the curve is notably narrower than a standard Gaussian, where the curve between the peak and the tails should be much smoother and less sharp.

Distribution of firing rates in Fig. 5 shows firing rate plotted against neuron activation, which is calculated as number of neurons with an average firing within a bin, further demonstrating inactivity of most excitatory neurons.

Fig. 6 shows the the coefficient of variation (CV) of the interspike intervals (ISI) of the excitatory neurons. The

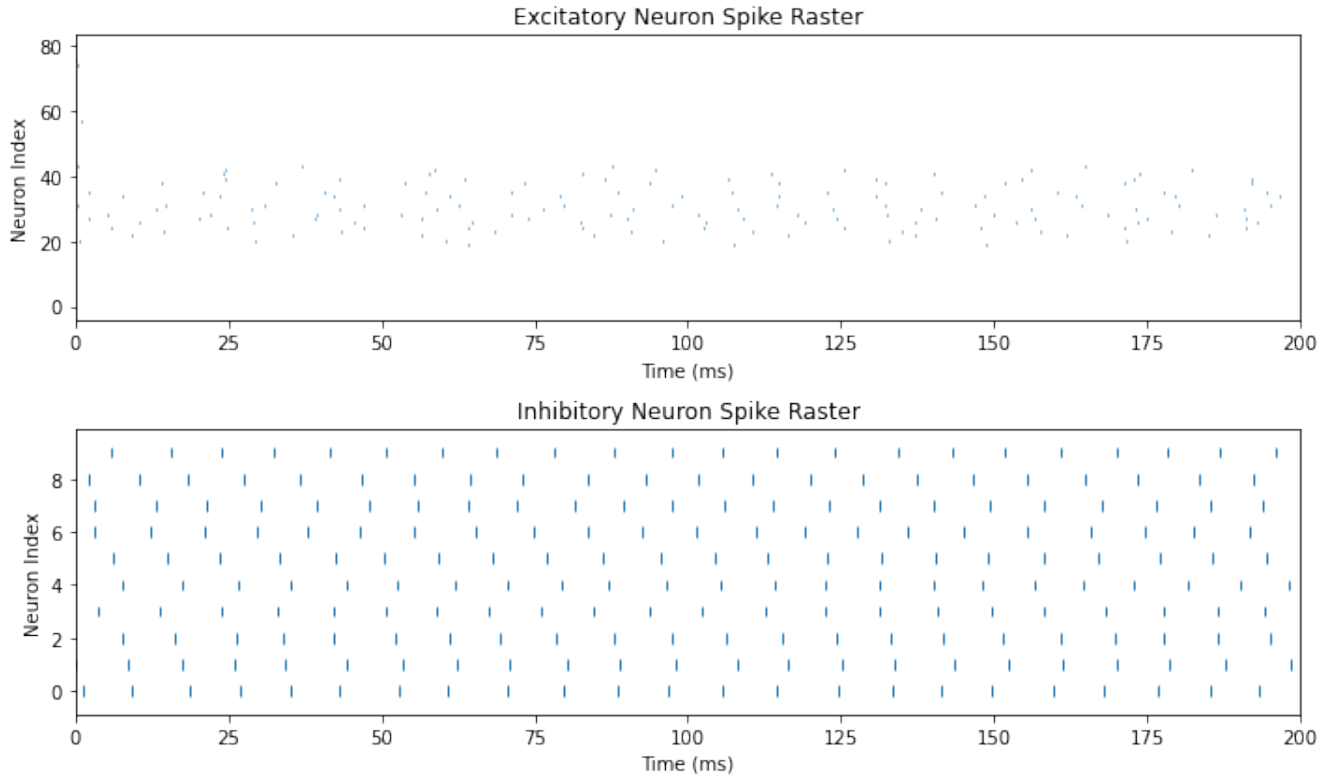


Figure 2. Sample spike train data from a network of 80 E and 10 I neurons, which is used for all subsequent figures.

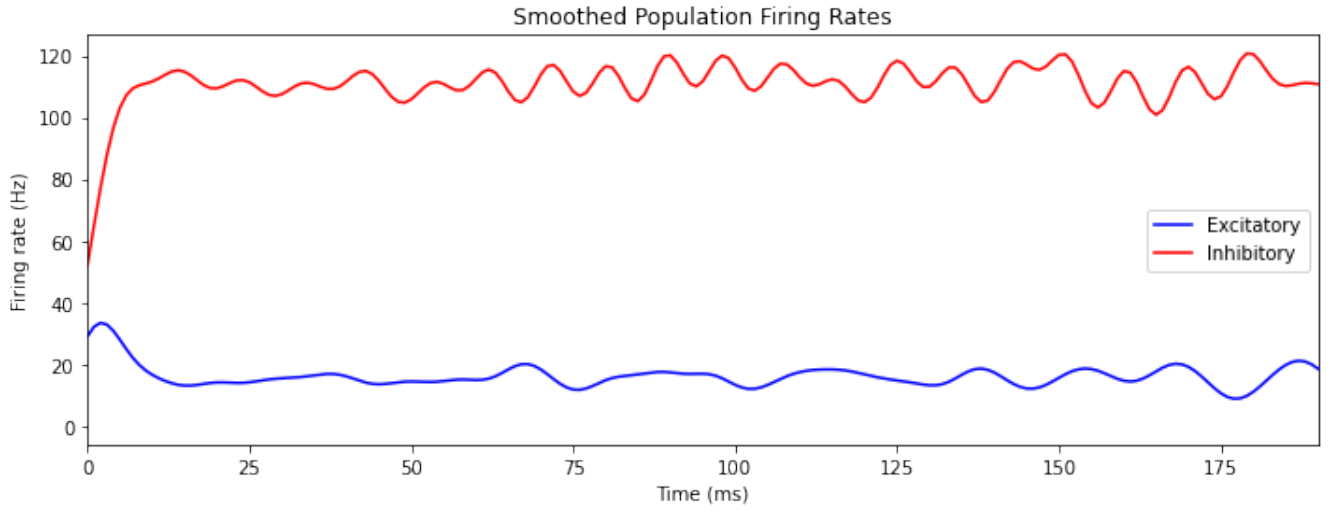


Figure 3. Smoothed PSTHs of spike trains shown in Figure 2.

CV serves as a measure of the irregularity in the firing of neurons, and aligns with the tuning curve, demonstrating the stochasticity introduced by adding 50% failure rates. The ISI distribution (Fig. 7) further complements this by showing the temporal patterns of spike events for each neuron. Neurons with short ISIs fire rapidly, implying a high firing rate, while

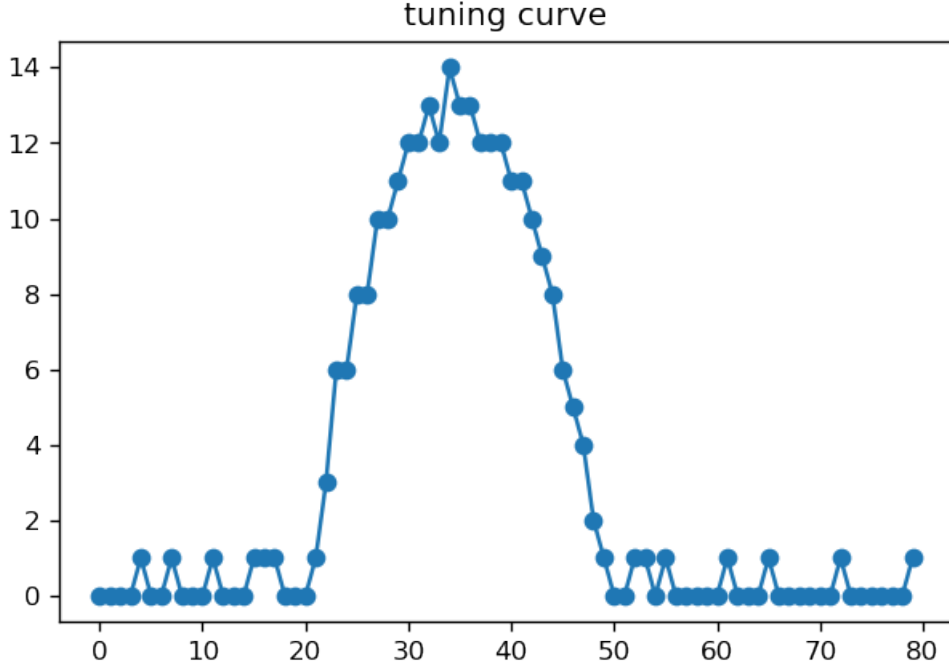


Figure 4. Tuning curves plotted as activity over neuron index.

those with long ISIs have slower or more sporadic firing patterns.

3.2 Auditory Tuning

To adapt this network to auditory tuning, we adjust the ring to accommodate multiple bumps, and introduce multiple cues, one weaker and one stronger, allowing for the initial network state to settle in different bump locations on the ring, with adjusted weights shown in Fig. 8. Interestingly, inhibition is shut down in this regime, yet a self-stable bump is maintained purely through excitatory drive. While multiple bumps did not appear, the tuning curve broadened, indicating that one cue likely overpowered the others and the network drifted into that state, in a winner-takes-all regime. In fact, this resembles the HD system more closely, which is proposed to have this winner-take-all regime when given one sharper, stronger cue and another broader, weaker cue. While this is unlikely in the auditory cortex, the overall regime sets up a broadened tuning curve with more overall neuron activation, rendering a closer fit to the Gaussian that is biophysically observed. Further fine-tuning of the model, including adjusting inhibition parameters could yield the separation of this single, wider bump into two distinct bumps.

Further work could include investigating correlation between network size and drift time, to investigate how quickly a self-sustaining bump can form with more stochasticity. In addition, connecting an excitatory and inhibitory ring together may yield the formation of multiple bumps as is likely in the auditory cortex. In addition, generating cues to be on the same location in a different pitch, methods from harmonic analysis could be used to distinguish the two signals, to differentiate between two bumps that share a tuning profile.

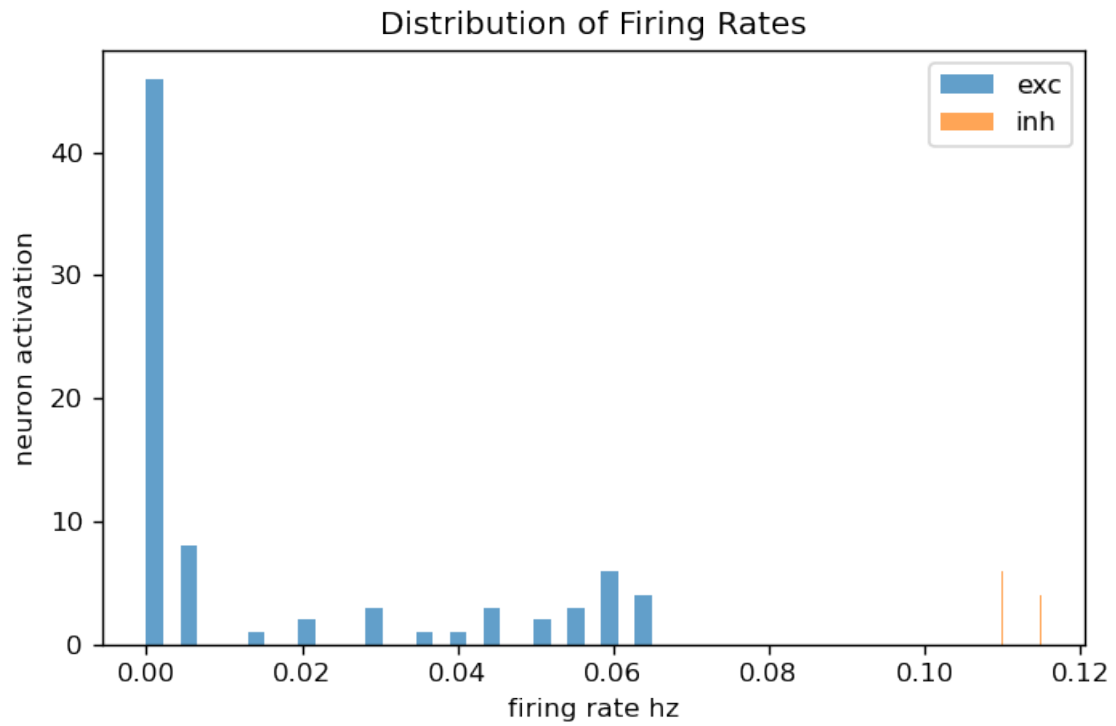


Figure 5. Distribution of firing rates.

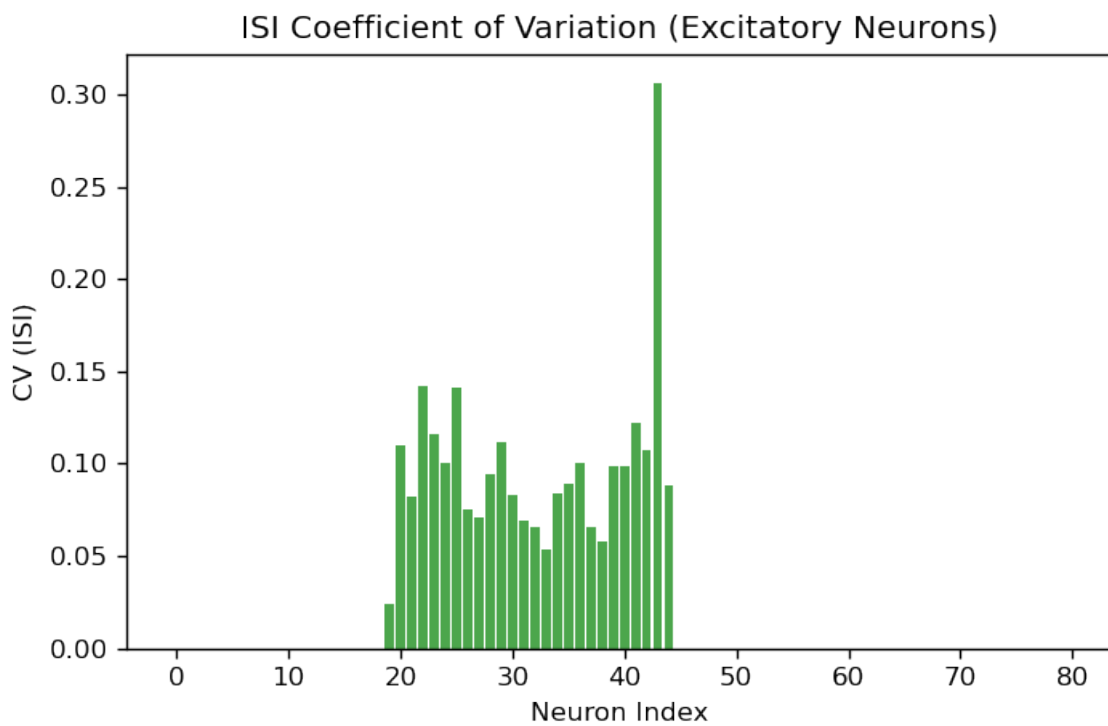


Figure 6. Coefficient of variation generated using ISIs for excitatory neurons by neuron index.

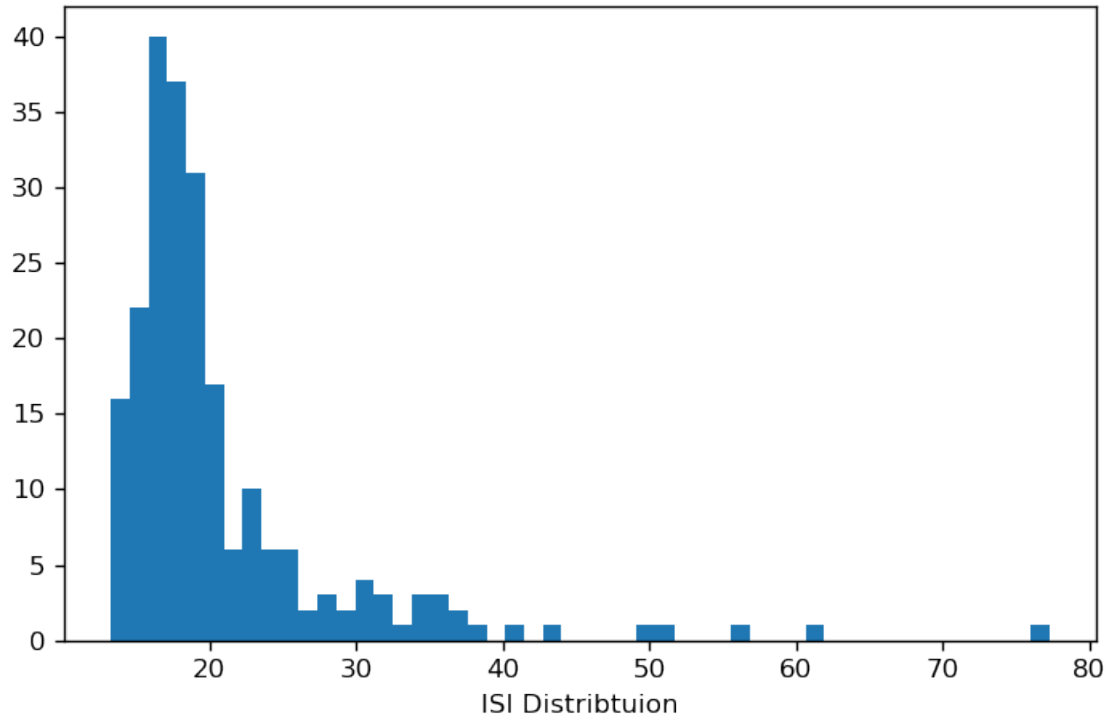


Figure 7. ISI distribution in bins of 10 ms between all ISIs between spikes for the simulation.

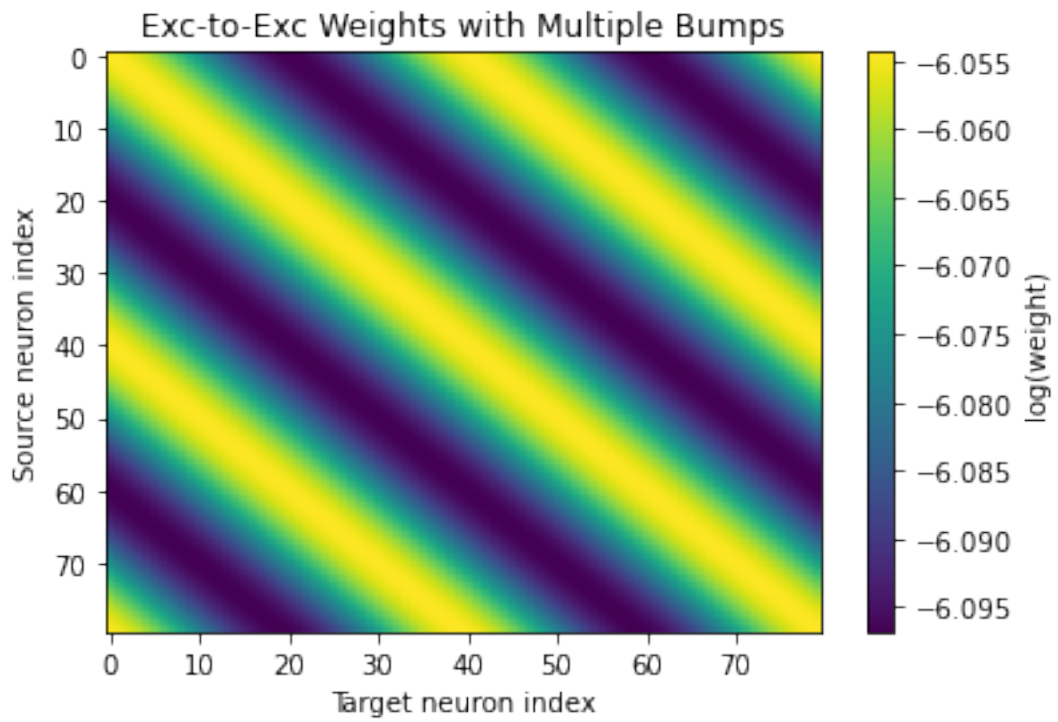


Figure 8. Updated weights matrix for ring with multiple bump locations.

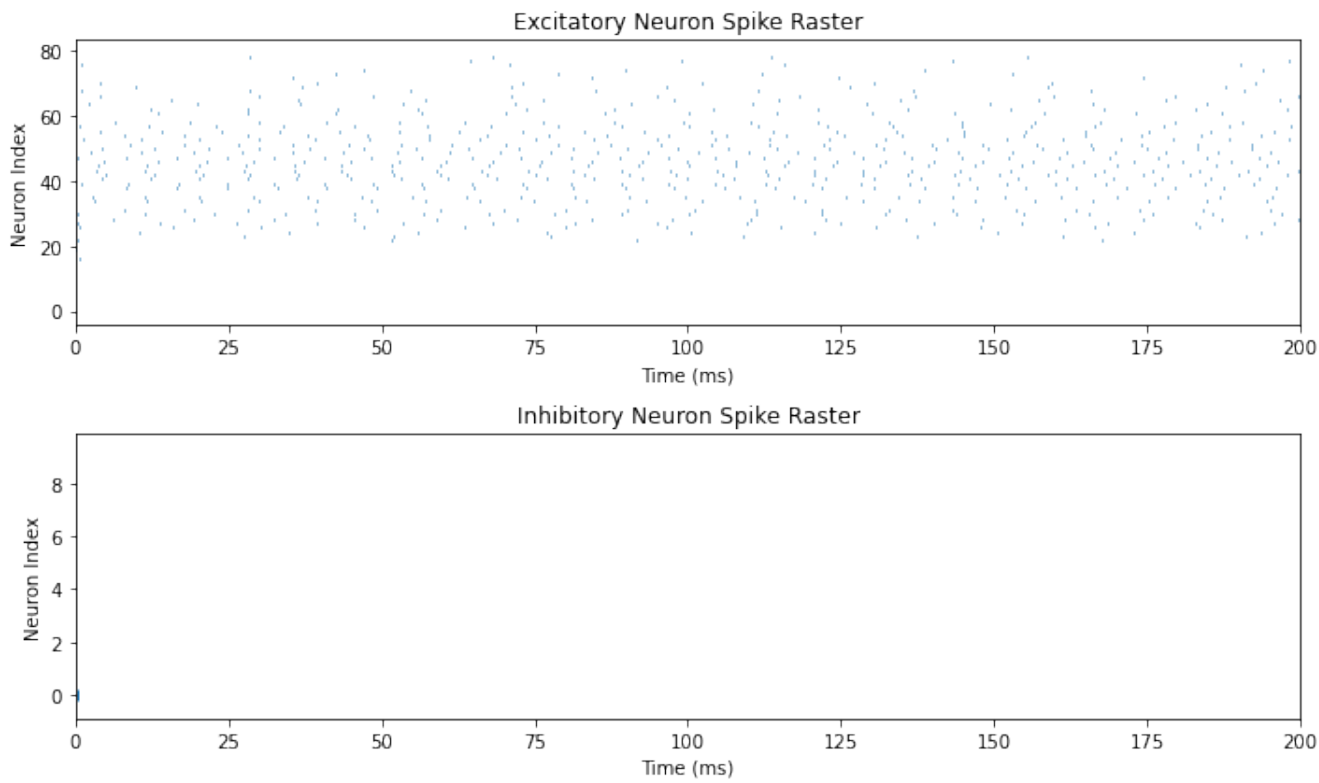


Figure 9. Sample spike train for updated network.

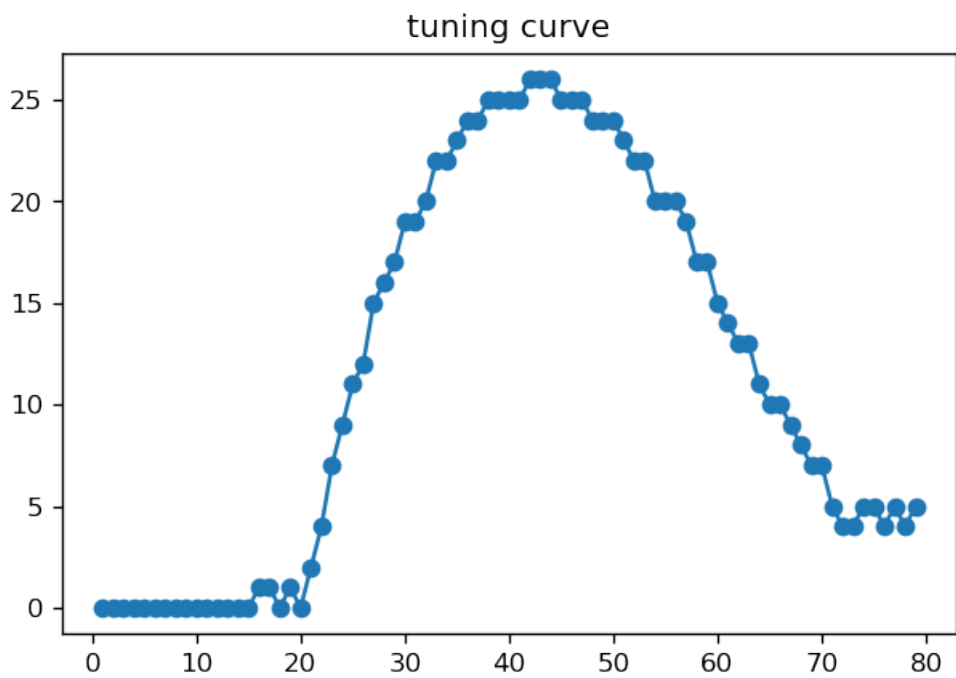


Figure 11. Broader tuning curve for ring with adjusted weights.

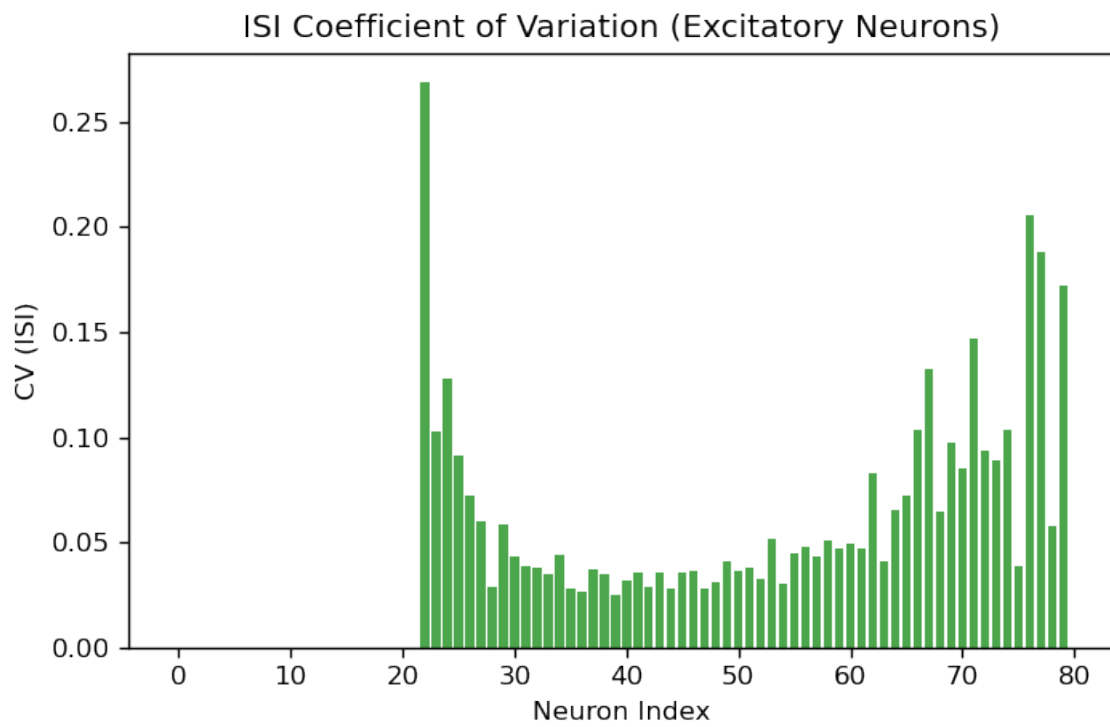


Figure 12. CV for all excitatory neurons with adjusted weights.